

Applied Materials – Drift-Sense: Sub-Pixel Pattern Matching & Mechatronic Stage Drift Compensation for SEM Wafer Metrology

Team Details

Team Name:

Signal Sense

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Solution Summary

A mechatronic-aware navigation recovery engine that combines multi-dimensional scale-rotation affine template matching with a center-bias decision prior and sub-pixel interpolation to achieve high-precision alignment of 100x reference patterns inside noisy 10x wide search frames

Mechanical Stage Drift Creates Micron-Scale Navigation Errors

The Semiconductor Metrology Challenge

- Wafer inspection tools must repeatedly revisit precise locations on multiple identical dies across the wafer surface.
- Mechanical stage vibrations, thermal expansion, and motor drift displace the physical landing coordinate away from the target feature.
- The problem requires aligning a 100x high-resolution reference pattern (1 nm/px) inside a larger, noisier 10x wide search image (10 nm/px).

Core Mechatronic Parameters

Reference Frame

1000 x 1000 px | 100x Zoom | 1 nm per pixel

Wide Search Frame

1000 x 1000 px | 10x Zoom | 10 nm per pixel

Magnification Shift

Nominal 10:1 (Robustness test: 9:1 to 11:1)

Rotational Offset

Minor mechanical drift of about 1 to 2 degrees

Coordinate Origin

Top-left (0,0) pixel convention

Four-Step Workflow Resolves 10:1 Scale Discrepancies and Stage Rotations

1. Data Ingestion

Procedurally loads 1000x1000 grayscale reference & search image pair.

Outcome: Identifies structure style

2. Template Scaling

Scales the reference image by 10x (downsampling) to match search pixel scale.

Outcome: Prepares search space

3. Affine Search

Iterates over rotation angles (-3° to $+3^{\circ}$) & scale factors (0.09 to 0.11) to compute normalized cross-correlation maps.

Outcome: Locates candidate peaks

4. Center Bias

Filters candidate peaks and selects the match closest to search frame center (500, 500) to survive period grid ambiguities.

Outcome: Outputs true (x, y)

DRAM Circles and FinFET Lines Pattern

Globally Unique Local Signatures

DRAM-Style Arrays

- DRAM layers consist of massive arrays of repeating circular contact holes.
- We procedurally generate dense matrices of circular contacts, applying minor randomized coordinate offsets to mimic lithographic variances.
- To provide a globally unique mechatronic signature, we introduce programmed defects such as random missing contacts and altered feature radii.

FinFET-Style Interconnects

- FinFET layers consist of dense, rectilinear, criss-crossing line channels (90° Manhattan geometry).
- We procedurally draw long horizontal and vertical lines with precise, randomized channel widths and spacing intervals.
- Random vertical line cuts and horizontal bridge patterns are dynamically generated to form a unique local fingerprint.

Physics-Inspired SEM Models Simulate Edge-Brightening and Streaking Disturbances

Edge-Brightening Effect

Scanning Electron Microscopes exhibit bright borders along steep topography steps due to enhanced secondary electron emission. We model this via morphological gradient filters.

Asymmetric Sensor Noise

Real SEM captures undergo Gaussian thermal blur, Poisson shot noise, and speckle noise. We inject randomized noise to the search and reference channels independently.

Charging Scanline Streaks

Local surface charge accumulation on insulating semiconductor substrates causes bright scanline streaks. We overlay random horizontal white intensity lines with Gaussian decay.

Stage Rotation and Scale Drift

To mimic real mechanical and scaling drift, we apply parameterized translation ($\pm 80\text{nm}$), angular rotation ($\pm 3^\circ$), and magnification drift (9:1 to 11:1 scale shifts).

Scale-Search and Center-Bias Filter Out Periodic Grating Ambiguities

Scale & Rotation Search Space

- Magnification limits vary between 9:1 and 11:1 due to scaling drift.
- We build a scale search space by resizing the reference template from 0.09 to 0.11 in steps of 0.005, preventing alignment failures.
- To survive stage rotations, we evaluate template orientations from -3° to $+3^\circ$ in steps of 1.5° .

The Center-Bias Decision Prior

- Highly repetitive and symmetric features (e.g. DRAM grids) yield multiple cross-correlation peaks of identical magnitude.
- Simple peak extraction risks choosing arbitrary adjacent periods, causing huge mechatronic coordinate errors.
- Our solver isolates all local peaks exceeding 95% of the absolute maximum and applies a mechatronic center-bias filter.
- The coordinate minimizing Euclidean distance to the image center (500, 500) is returned as the predicted target center.

Decoupled Standalone Scripts Run Ingestion and Inference in Three Commands

CLI Commands for Execution

```
# Install dependencies
pip install -r requirements.txt

# Generate the 30-pair synthetic benchmark
python3 generate_dataset.py --style DRAM \
    --num_pairs 30 --out_dir ./test_dataset

# Localize any reference inside wide search
python3 localize.py ./test_dataset/ref_0.png \
    ./test_dataset/search_0.png
```

Standalone Submissions Structure

generate_dataset.py

Procedural generator script that outputs paired 1000x1000 grayscale reference and wide search images along with true coordinate labels.

localize.py

Production-ready inference script running scale-and-rotation template localization. Returns predicted (x, y) coordinates closest to the center.

test_drift_sense.py

Benchmarking script that executes both generator and localizer over 30 test pairs, calculating accuracy pass rates.

requirements.txt

Condensed environment profile outlining dependencies like numpy and opencv-python.

Independently Generated 30-Pair Dataset Evaluates Algorithmic Robustness

Benchmark Composition

- A complete, independent test set of 30 distinct image pairs (15 DRAM, 15 FinFET) is procedurally generated with varying coordinates.
- Search coordinates are dynamically generated and recorded as ground-truth labels for absolute validation.
- We inject independent sensor noise profiles on reference and search images to ensure zero leak of noise patterns.
- Stage drift transformations are parameterized to evaluate behavior under scaling, rotation, and translation boundaries.

Validation Search Space Parameters

DRAM Features: Contact diameter: 50nm | Inter-cell pitch: 100nm

FinFET Features: Line width: 40nm | Channel pitch: 110nm

Noise Settings: Speckle noise variance: 0.1 | Horizontal charging scanlines: 3

Rotation Bounds: Angle offset: $\pm 3.0^\circ$ | Incremental search: 1.5°

Scale Bounds: Zoom limits: 0.09 to 0.11 | Step size: 0.005

Proposed Algorithm Achieves 96.67% Pass Rate Under 5-Pixel Tolerance

Benchmark Statistics

Evaluated Cases: 30 randomized pairs

Mean Euclidean Error: 4.879 pixels

Median Euclidean Error: 1.222 pixels

Worst-Case Error: 110.166 pixels

Mean Runtime / Pair: 1245.9 milliseconds

Threshold-Wise Pass Rates

Pass Rate @ 5-pixel threshold

96.67% (29 / 30 cases)

Pass Rate @ 4-pixel threshold

96.67% (29 / 30 cases)

Pass Rate @ 2-pixel threshold

86.67% (26 / 30 cases)

Pass Rate @ 1-pixel threshold

33.33% (10 / 30 cases)

Mean DRAM Alignment Accuracy

0.74 pixels (Sub-pixel repeat)

Rotation Search Outperforms Standard Matchers by Suppressing Periodic False Peaks

Limitations of Standard Matchers

- Traditional template matching has no intrinsic scale or rotation invariance. Rotations $>1.0^\circ$ drop similarity score below critical thresholds.
- Standard algorithms search on exact magnification ratios. If the system undergoes zoom drift (9:1 or 11:1), localization completely fails.
- Without center-bias priority, standard matchers pick arbitrary matching periods in high-frequency arrays, leading to colossal coordinate shifts.

Strengths of the Drift-Sense Solver

- Robust Scale and Angle Scanning survives off-nominal magnification drift and mechanical stage rotations.
- Dynamic Peak Extraction identifies all matching periodic regions instead of stopping at the absolute global maximum.
- Mechatronic Center-Bias selects the coordinate closest to search frame center, matching physical stage expectations.
- Achieves 96.67% pass rate where standard baseline matchers score below 20% on the same dataset.

Periodic Grid Spacing of 110 Pixels Triggers Phase-Wrapping Grid-Ambiguity

Analysis of FinFET Outlier Case

- In FinFET Pair 16, our algorithm returned an error of exactly 110.17 pixels.
- FinFET features are highly periodic parallel lines with a repeating channel spacing of exactly 110 pixels.
- Under extreme charging scanlines and speckle noise, the edge-brightening profiles of the target channel degraded.
- The cross-correlation maximum of the adjacent line shifted slightly higher than the target line, trapping the algorithm on a parallel grid channel.

Algorithmic Mitigation

- A purely visual algorithm cannot resolve absolute grid shift in highly uniform vertical lines without global context.
- Mitigation 1: Design-Integrated Care Area Mapping. Align physical stage coordinates directly to design CAD layouts (such as GDSII or OASIS layouts) to select locally unique landmarks.
- Mitigation 2: Multi-Sensor Fusion. Combine secondary electron scanning with backscattered electron images to capture underlying material layers, breaking visual symmetry.

Three Engineering Upgrades Define the Future CAD-Integrated Roadmap

CAD-to-SEM Mapping

GDSII / OASIS layouts can be aligned to design care areas, allowing the tool to avoid non-unique repeating arrays.

Status: Planned Integration

Multi-Detector Fusion

Fuse secondary electron (SE) signals with backscattered electron (BSE) sensors to resolve vertical layers.

Status: Researching Sensors

Inference Optimization

Implement GPU-accelerated PyTorch/ONNX-based solvers to reduce inference latency from 1.2s to <100ms.

Status: Prototyping Code